

## How the Proof Works

We start with a smooth, incompressible flow of finite energy and keep its viscosity fixed and positive. Its velocity  $u$  and pressure  $p$  satisfy the three-dimensional Navier–Stokes equation:

$$\underbrace{\partial_t u}_{\text{local acceleration}} + \underbrace{(u \cdot \nabla)u}_{\text{nonlinear transport}} = \underbrace{-\nabla p}_{\text{pressure}} + \underbrace{\nu \Delta u}_{\text{viscous diffusion}}, \quad \nabla \cdot u = 0.$$

We know it stays smooth for a short time; the question is whether it stays smooth forever or becomes singular in finite time. The hope is simple: viscosity smooths the fluid faster than nonlinear transport can sharpen it.

Consider the case of a vortex within the fluid. Its vorticity  $\omega = \nabla \times u$  satisfies

$$\partial_t \omega + (u \cdot \nabla) \omega = (\omega \cdot \nabla) u + \nu \Delta \omega.$$

Strain within the fluid's flow can naturally stretch a vortex. When that happens, incompressibility forces a material tube to narrow as it lengthens. For a thin tube that remains locally circular, a positive axial strain  $\sigma$  increases its length  $\ell$  and reduces its radius  $r$ :

$$\dot{\ell} = \sigma \ell > 0, \quad r \propto \ell^{-1/2}.$$

The fluid now circulates around a smaller radius. If the enclosed circulation  $\Gamma$  is retained, its mean swirl speed  $v$  increases:

$$v = \frac{|\Gamma|}{2\pi r}.$$

But the tube follows a chosen group of fluid particles, while the vortex core is the region where rotation is concentrated. Stretching squeezes those particles inward. Viscosity diffuses momentum between neighbouring regions of fluid, reducing velocity differences and spreading the rotation through the surrounding fluid. Over an elapsed time  $\tau$ , its diffusion distance is of order

$$r_{\text{diff}} \sim \sqrt{\nu \tau}.$$

The material tube can narrow while some rotation spreads beyond its boundary, reducing the circulation enclosed by it. So narrowing makes the fluid circulate faster only if the enclosed circulation does not weaken too quickly. Whether viscosity can smooth those differences fast enough is the difficulty.

If this concentration kept going, the region of fast motion would become smaller and smaller. We would have to follow it toward vanishing width, where even very large velocities could occupy very little volume. Looking at the total energy would tell us less and less about what was happening inside that region. To compare transport and diffusion there, we rescale the flow: shrink lengths by  $\lambda > 1$ , increase velocities by  $\lambda$ , and shorten times by  $\lambda^2$ :

$$u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t),$$

$$p_\lambda(x, t) = \lambda^2 p(\lambda x, \lambda^2 t).$$

Every term in the momentum equation acquires a factor  $\lambda^3$ . Removing it leaves the original equation with the viscosity still fixed at  $\nu$ . A narrower, faster flow has preserved the scaling balance between transport and diffusion. Shrinking the region has given viscosity no automatic advantage; their actual competition still depends on how the flow is arranged.

We need a measurement that can see the spatial variation inside this shrinking region. The Sobolev index  $s$  determines how strongly we count fine detail. At integer values it counts spatial derivatives in  $L^2$ ; fractional values fill in between them by weighting frequency  $\xi$  by  $|\xi|^s$ . Under rescaling, velocity contributes  $\lambda$ , the derivative weight contributes  $\lambda^s$ , and the square root of the volume contributes  $\lambda^{-3/2}$ . Thus

$$\|u_\lambda(\cdot, t)\|_{\dot{H}^s} = \lambda^{s-1/2} \|u(\cdot, \lambda^2 t)\|_{\dot{H}^s}.$$

Below  $s = \frac{1}{2}$ , concentration makes the norm smaller. Above  $s = \frac{1}{2}$ , it makes the norm larger. Exactly at  $s = \frac{1}{2}$ , the finer variation offsets the shrinking volume and the norm is unchanged. This is the critical height: the measurement retains its sensitivity as we change scale. That invariance compares rescaled flows; it does not say the norm stays constant as one flow evolves.

Energy sits below that height. With  $E = \frac{1}{2} \|u\|_2^2$ ,

$$E[u_\lambda](t) = \lambda^{-1} E[u](\lambda^2 t).$$

The velocity grows, yet the volume containing it shrinks fast enough for the total energy to fall. An energy bound can coexist with increasingly fast motion in an increasingly narrow region. It cannot, by itself, stop this concentration.

The time needed for each further stage also gets shorter. At width  $r$  and velocity scale  $U$ , transport acts over a time of order  $r/U$ , while viscosity acts over a time of order  $r^2/\nu$ . Both are shortened by  $\lambda^2$  under the rescaling. If a flow could repeat such stages, their durations would fit into a finite total:

$$\tau_j = \lambda^{-2j} \tau_0, \quad \sum_{j=0}^{\infty} \tau_j = \frac{\tau_0}{1 - \lambda^{-2}} < \infty.$$

This is the hypothetical Zeno cascade. Infinitely many narrowing stages could accumulate before a finite time  $T_*$ , with no smallest width and no final stage of acceleration. The scaling permits that picture; whether the fluid can actually carry it out is the regularity problem.

For a finite-time singularity, velocity would have to blow up. That requires acceleration to blow up, then jerk, and so on, and so on. To see the necessity, follow the velocity field at fixed spatial positions. Its temporal responses are  $u, \partial_t u, \partial_t^2 u, \dots$ , and for any  $t_0 < T_*$ ,

$$\|\partial_t^j u(t)\|_\infty \leq \|\partial_t^j u(t_0)\|_\infty + \int_{t_0}^t \|\partial_t^{j+1} u(s)\|_\infty ds.$$

A bounded next derivative could produce only a finite change in the time remaining. Unbounded velocity requires an infinite accumulated temporal response, which forces that response to become unbounded too. Apply this again and the requirement reaches every finite temporal order. These are Eulerian responses; the acceleration of a moving fluid particle is  $D_t u = \partial_t u + (u \cdot \nabla) u$ . The claim concerns unbounded spatial norms on the approach to  $T_*$ , without requiring monotone growth or growth at one fixed point.

Those increasingly demanding temporal responses are generated by the Navier–Stokes equation itself. Differentiating it lets us see what the fluid would have to supply at each order:

$$\begin{aligned}\partial_t u &= \nu \Delta u - (u \cdot \nabla)u - \nabla p, \\ \partial_t^2 u &= \nu \Delta \partial_t u - (\partial_t u \cdot \nabla)u \\ &\quad - (u \cdot \nabla) \partial_t u - \nabla \partial_t p, \\ \partial_t^3 u &= \nu \Delta \partial_t^2 u - (\partial_t^2 u \cdot \nabla)u \\ &\quad - 2(\partial_t u \cdot \nabla) \partial_t u - (u \cdot \nabla) \partial_t^2 u - \nabla \partial_t^2 p.\end{aligned}$$

Each differentiation splits the transport product again, while viscosity applies a Laplacian to the preceding response. The second equation contains  $\Delta \partial_t u$ ; substituting the first equation into it exposes  $\Delta^2 u$ . The next substitution exposes  $\Delta^3 u$ . Higher time derivatives keep reaching further into the spatial structure of the flow. The complete pattern retains all the accompanying nonlinear terms:

$$\begin{aligned}\partial_t^{n+1} u &= \nu \Delta \partial_t^n u \\ &\quad - \sum_{a=0}^n \binom{n}{a} (\partial_t^a u \cdot \nabla) \partial_t^{n-a} u - \nabla \partial_t^n p.\end{aligned}$$

We write these velocity and pressure responses as  $V_n = \partial_t^n u$  and  $Q_n = \partial_t^n p$ .

At the critical height, this temporal-to-spatial relation becomes a relation between Sobolev energies. Write  $E_s = \frac{1}{2} \|\Lambda^s u\|_2^2$ , where  $\Lambda = (-\Delta)^{1/2}$ , and let  $H = E_{1/2}$ . Differentiating  $E_s$  pairs the velocity with its temporal response. The viscous Laplacian moves one derivative onto each factor under integration by parts, exposing the next spatial height:

$$E'_s = -2\nu E_{s+1} + \mathcal{P}_s,$$

where the transport and pressure contribution is retained as

$$\mathcal{P}_s = -\operatorname{Re} \langle \Lambda^s u, \Lambda^s ((u \cdot \nabla)u + \nabla p) \rangle_{L^2}.$$

Differentiate  $H$  once and  $E_{3/2}$  appears. Differentiate again and the changing  $E_{3/2}$  exposes  $E_{5/2}$ :

$$\begin{aligned}H' &= -2\nu E_{3/2} + \mathcal{P}_{1/2}, \\ H'' &= 4\nu^2 E_{5/2} - 2\nu \mathcal{P}_{3/2} + \partial_t \mathcal{P}_{1/2}.\end{aligned}$$

Continuing through  $H'''$ ,  $H^{(4)}$ ,  $\dots$  exposes  $E_{7/2}$ ,  $E_{9/2}$ ,  $\dots$ , together with the differentiated nonlinear contributions. The temporal sequence that a singularity would force us to confront has led directly to an unending sequence of spatial Sobolev heights.

And those heights are exactly what smoothness needs. For any chosen number  $k$  of continuous spatial derivatives, Sobolev embedding asks for a finite height above  $k + \frac{3}{2}$ :

$$m > k + \frac{3}{2} \implies H^m(\mathbb{R}^3) \hookrightarrow C^k(\mathbb{R}^3).$$

If every finite height is available, there is no last derivative we can ask for. The intersection contains enough regularity for every choice of  $k$ :

$$L^2 \cap \bigcap_{n \geq 0} \dot{H}^{n+1/2} = \bigcap_{m \geq 0} H^m = H^\infty \hookrightarrow C^\infty.$$

This is where the temporal and spatial demands meet. The equation connects each new temporal response to higher spatial derivatives, and the intersection of all those spatial requirements is smoothness. With those requirements controlled together up to  $T_*$ , the limiting velocity would be smooth enough to start the flow again. The identities expose that possibility; realizing it requires bounds on the full approach to  $T_*$ , including the nonlinear contributions and agreement between all the retained derivatives.

If a singularity were to form, we would expect the opposite behavior. The flow could remain smooth at every earlier instant while some fixed spatial Sobolev norm became unbounded on the approach to  $T_*$ . Its temporal responses would become unbounded too, even while total energy remained finite. Every shorter interval could contain all finite derivatives, yet their control would deteriorate as its endpoint approached the singular time. We would lose the regularity needed to carry the flow through  $T_*$ . That is the loss the construction has to exclude.

We begin that construction with the initial velocity  $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$ : every finite spatial Sobolev order is available at the start, and the velocity is incompressible. Incompressibility fixes pressure through  $-\Delta p = \partial_i \partial_j (u_i u_j)$ , with its decay normalization. Differentiating this relation along with the velocity determines the pressure response at each order. In the notation already introduced,

$$V_{n+1} = \nu \Delta V_n - \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} - \nabla Q_n,$$

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i} V_{n-a,j}.$$

Here  $R_i$  are the Riesz transforms and repeated spatial indices are summed. At  $t = 0$ ,  $V_0 = u_0$  fixes the first response, which fixes the next, and so on. Spatial differentiation fills out each temporal row. These velocity and pressure derivatives form the mixed jet generated by the initial datum. Its entries remain connected through the equation that generated them.

Any finite regularity demand uses only a finite part of this structure. A temporal depth  $N$  and a spatial height  $M$  select a rectangle  $\mathcal{R}_{N,M}[u_0; T_*]$ . Each coordinate measures a temporal response  $V_n$  in a spatial Sobolev space; adjacent temporal rows retain  $\partial_t V_n = V_{n+1}$ , and the pressure rows accompany them. The datum-generated rectangle theorem, Theorem 4.6, controls this finite selection throughout the approach to  $T_*$ :

$$\sup_{0 < T < T_*} \sum_{n=0}^N \left( \|V_n\|_{L^2(0,T;H^{M+1})}^2 + \|V_{n+1}\|_{L^2(0,T;H^{M-1})}^2 \right) < \infty.$$

For fixed  $N, M$ , the bound stays finite as  $T$  approaches  $T_*$ . It may increase when we ask for more derivatives. What it excludes is divergence at any fixed pair of orders as the proposed singular time approaches.

The adjacent spaces let each retained response reach that time. The upper row  $V_n \in L^2(0, T_*; H^{M+1})$  and its temporal derivative  $V_{n+1} \in L^2(0, T_*; H^{M-1})$  pair across  $H^M$ . Their square-integrability makes the change in the squared  $H^M$  norm integrable. The Hilbert-triple trace theorem then gives

$$V_n \in C([0, T_*]; H^M).$$

So each finite rectangle supplies actual endpoint values at its retained orders. A larger rectangle keeps every derivative already present, and on an overlap both rectangles contain derivatives of the original velocity and pressure. Their entries agree, including their traces at  $T_*$ .

This agreement lets us retain every finite rectangle together. Increasing temporal depth and spatial height independently gives the compatible intersection

$$\mathcal{I}[u_0] := (\mathcal{R}_{N,M}[u_0; T_*])_{N,M \geq 0},$$

$$u \in \bigcap_{r,m \geq 0} H^r(0, T_*; H^m(\mathbb{R}^3)).$$

Every finite request for derivatives fits inside a sufficiently large rectangle, and every larger rectangle preserves the answer. We do not need one bound that works uniformly over all orders. Each finite order has its own bound, and compatibility makes them properties of one flow.

In particular, the zeroth temporal row reaches  $T_*$  in every spatial Sobolev space. Its traces agree as a single limiting velocity:

$$u_* = \lim_{t \uparrow T_*} u(t) \quad \text{in every } H^m,$$

$$u_* \in H_\sigma^\infty \hookrightarrow C^\infty.$$

The subscript  $\sigma$  records incompressibility. The differentiated equations also reach the endpoint, fixing its temporal responses from  $u_*$ . Smooth local existence starts a solution there, and uniqueness joins it to the original flow. The flow continues beyond its supposed maximal time. A finite  $T_*$  is impossible, so  $T_* = \infty$ .

With the intersection constructed, we can derive the finite estimates, and use them. For any finite time  $T$ , temporal order  $r$ , spatial order  $k$ , and  $2 \leq p \leq \infty$ , choose  $m > k + \frac{3}{2} - \frac{3}{p}$ . A sufficiently large rectangle and Sobolev embedding give

$$\partial_t^r u \in C([0, T]; H^m) \hookrightarrow L^\infty(0, T; W^{k,p}).$$

For example,  $r = 0$ ,  $k = 1$ ,  $p = \infty$ , and  $m = 3$  give

$$\int_0^T \|\nabla u(t)\|_\infty dt \leq T \sup_{0 \leq t \leq T} \|\nabla u(t)\|_\infty < \infty.$$

Return to the locally circular material tube at the start. Its axial strain satisfies  $\sigma \leq \|\nabla u\|_\infty$ , while incompressibility gives  $\dot{r}/r = -\sigma/2$ . Integrating,

$$r(t) \geq r(0) \exp\left(-\frac{1}{2} \int_0^t \|\nabla u(s)\|_\infty ds\right) > 0.$$

Only a finite amount of stretching can accumulate in a finite time. The tube cannot be squeezed to zero radius there.